EFY TESTED
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SMART HOME AUTOMATION SYSTEM Using ESP8266 NodeMCU

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IoT-based smart home automation lets homeowners control appliances conveniently from their smartphones and laptops. Appliances and lights frequently stay on unnecessarily, resulting in significant energy wastage.

Implementing home automation and smart technology can automate energy-saving routines, such as turning off lights or shutting down electronics when not in use. Smart home automation enables homeowners to monitor their appliances even when they are away.

Smart home technology can enhance overall quality of life by simplifying daily tasks and reducing stress. For example, automated lighting and blinds can create a more comfortable and pleasant living environment. The purpose of this system is illustrated in the block diagram shown in Fig. 1. Components used in the automation system are listed in the bill of materials Table 1.

Circuit diagram

The circuit of the smart home automation system is shown in Fig. 2. It is built around ESP8266 (MOD1), a four-channel relay module (MOD2), and a few jumper wires. A 5V battery/power bank is used to power the system. Connections between ESP8266 and relay inputs are given in Table 2, while connections between the relay module and load (bulb or fan) are given in Table 3.

Dashboard design using Blynk Cloud

Now that we have all the components, we can start making our IoT-based smart home automation system—Dashboard design using Blynk Cloud. First, create BLYNK_TEMPLATE_ID, BLYNK_TEMPLATE_NAME, and BLYNK_AUTH_ID by following these steps:

Step 1. Visit *https://blynk.io/* and complete your registration process. You will receive a user ID and password.

Step 2. Log in to the Blynk Cloud from your laptop using the generated user ID and password. Click on 'create new template' option, then select 'web dashboard.'

TABLE 1
BILL OF MATERIALS

Component Name	Quantity	Description
ESP8266 (MOD1)	1	For programming
Relay Module 4 Channel (MOD2)	1	To interface AC operated load
Male to female jumper wires	10	For connection
5V battery/power bank	1	Power supply

TABLE 2
CONNECTIONS BETWEEN ESP8266 AND RELAY MODULE

ESP8266 (MOD1)	Components and Pin
ESP8266 Pin D1	IN0 (Channel 0 of relay module)
ESP8266 Pin D2	IN1 (Channel 1 of relay module)
ESP8266 Pin D5	IN2 (Channel 2 of relay module)
ESP8266 Pin D6	IN3 (Channel 3 of relay module)
VCC of ESP8266	VCC of relay module
GND of ESP8266	GND of relay module

Fig. 1: Block diagram of the project

TABLE 3
RELAY AND LOAD (BULB)

Relay	Description
Pole of relay module	Connected to phase line of 230V AC supply
N/O (normally open) pin of relay module	Connected to one terminal of each lamp
N/C (normally closed) pin of relay module	Neutral line of 230V AC connected to second terminal of each lamp